

NMEA2000 실습

2017. 5

한국선박전자산업진흥협회

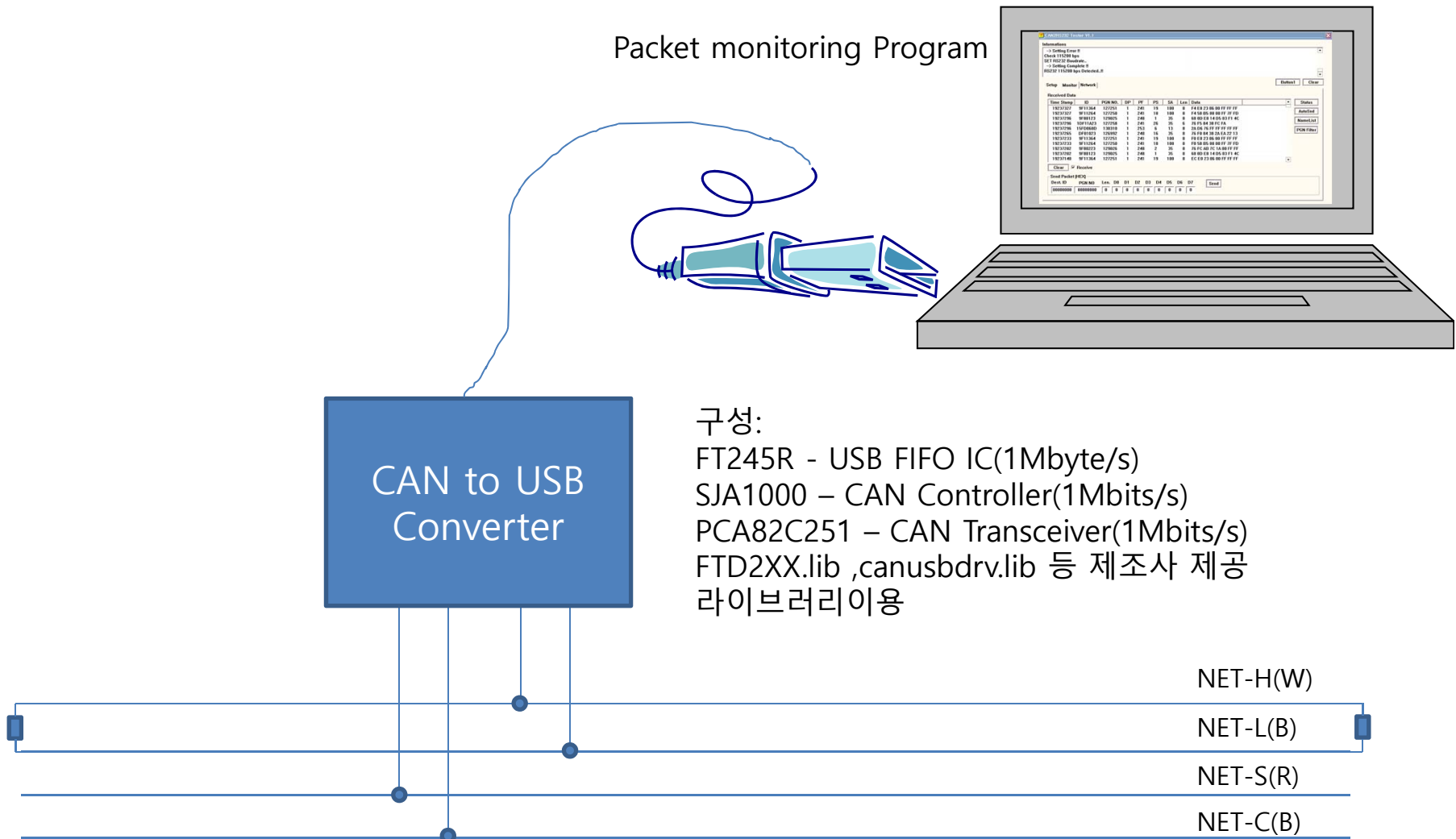
조선해양 ICT융합연구소(OSICT)

MEIPA(Marine Electronics Industry Promotion
Association)

실습목표

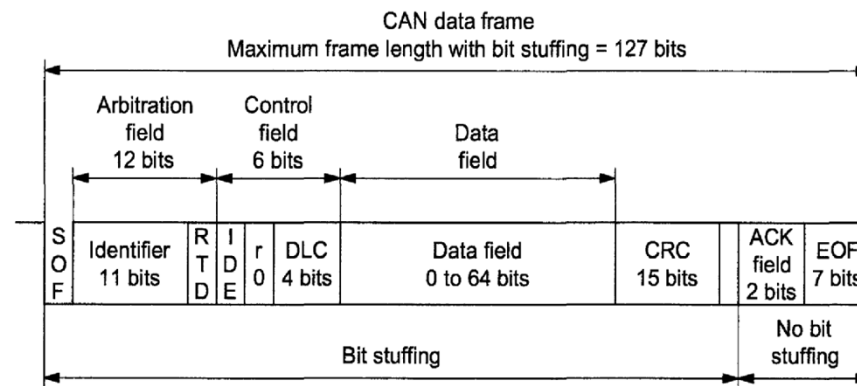
- 1. NMEA2000의 Data Packet 이해
- 2. Single Packet 분석
- 3. ISO Request PGN 59904 실습
- 4. ISO Address Claim PGN 60928 분석
- 5. Address Claiming 실습

1. NMEA2000의 Data Packet 이해

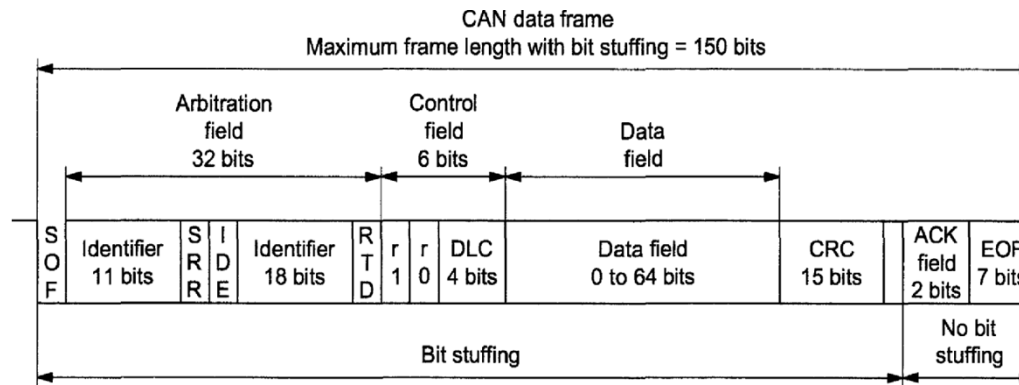


1. NMEA2000의 Data Packet 이해

CAN frame format



a) CAN base frame format

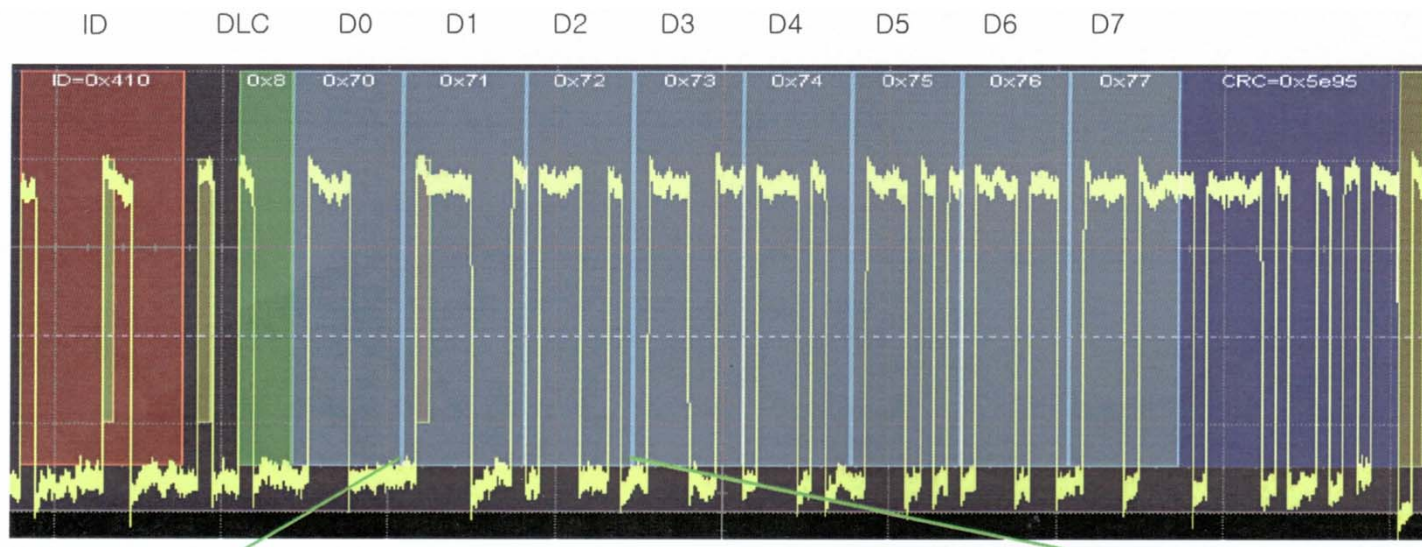


b) CAN extended frame format

Figure 1 — CAN data frames

1. NMEA2000의 Data Packet 이해

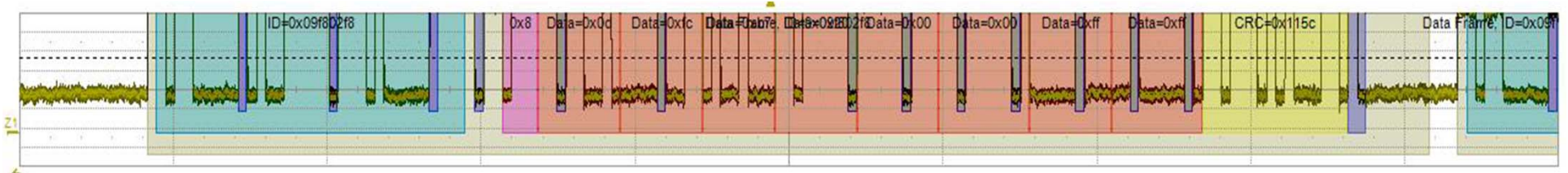
예제) CAN Base Frame Format



Item	Value
Identifier (11bits)	
DLC	
Data(8bytes/64bits)	

1. NMEA2000의 Data Packet 이해

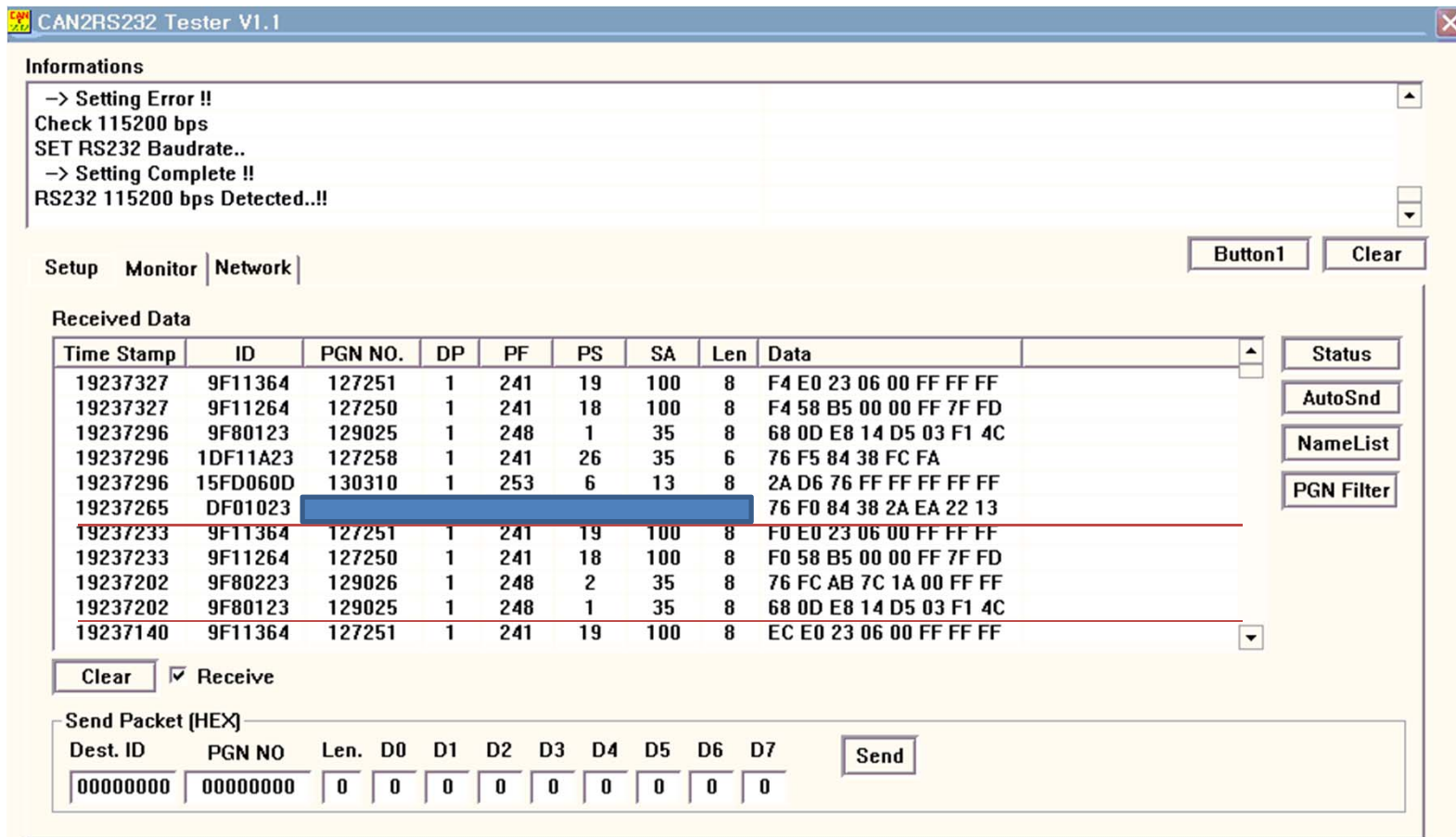
실습과제) CAN Extended Frame Format



Item	Value
Identifier (29bits)	
DLC	
Data(8bytes/64bits)	

2. Single Packet 분석

예제) PGN 129025 싱글 패킷 분석



2. Single Packet 분석

ID 분석 : 9F80123

PDU Format	29 bit Identifier						PGN 개수
	Priority 3bit	EDP 1bit	DP 1bit	PF 8bit	PS 8bit	SA 8bit	
		PGN					
PDU1	0-7	0	0	0-239	DA 0-255	SA 0-255	240
			1	0-239	DA 0-255	SA 0-255	240
PDU2	0-7	0	0	240-255	GE 0-255	SA 0-255	16x256 =4096
			1	240-255	GE 0-255	SA 0-255	16x256 =4096

PDU2:

Priority			EDP	DP	PF								PS								SA							
0	1	0	0	1	1	1	1	1	1	0	0	0	0	0	0	0	0	1	0	0	1	0	0	0	1	1		
			0111111000000000001B = <u>129025</u>																									

2. Single Packet 분석

PGN 데이터 분석

Position, Rapid Update **PGN: 129025**
hex: 1F801

This PGN provides latitude and longitude referenced to WGS84. Being defined as single frame message, as opposed to other PGNs that include latitude and longitude and are defined as fast or multi-packet, this PGN lends itself to being transmitted more frequently without using up excessive bandwidth on the bus for the benefit of receiving equipment that may require rapid position updates.

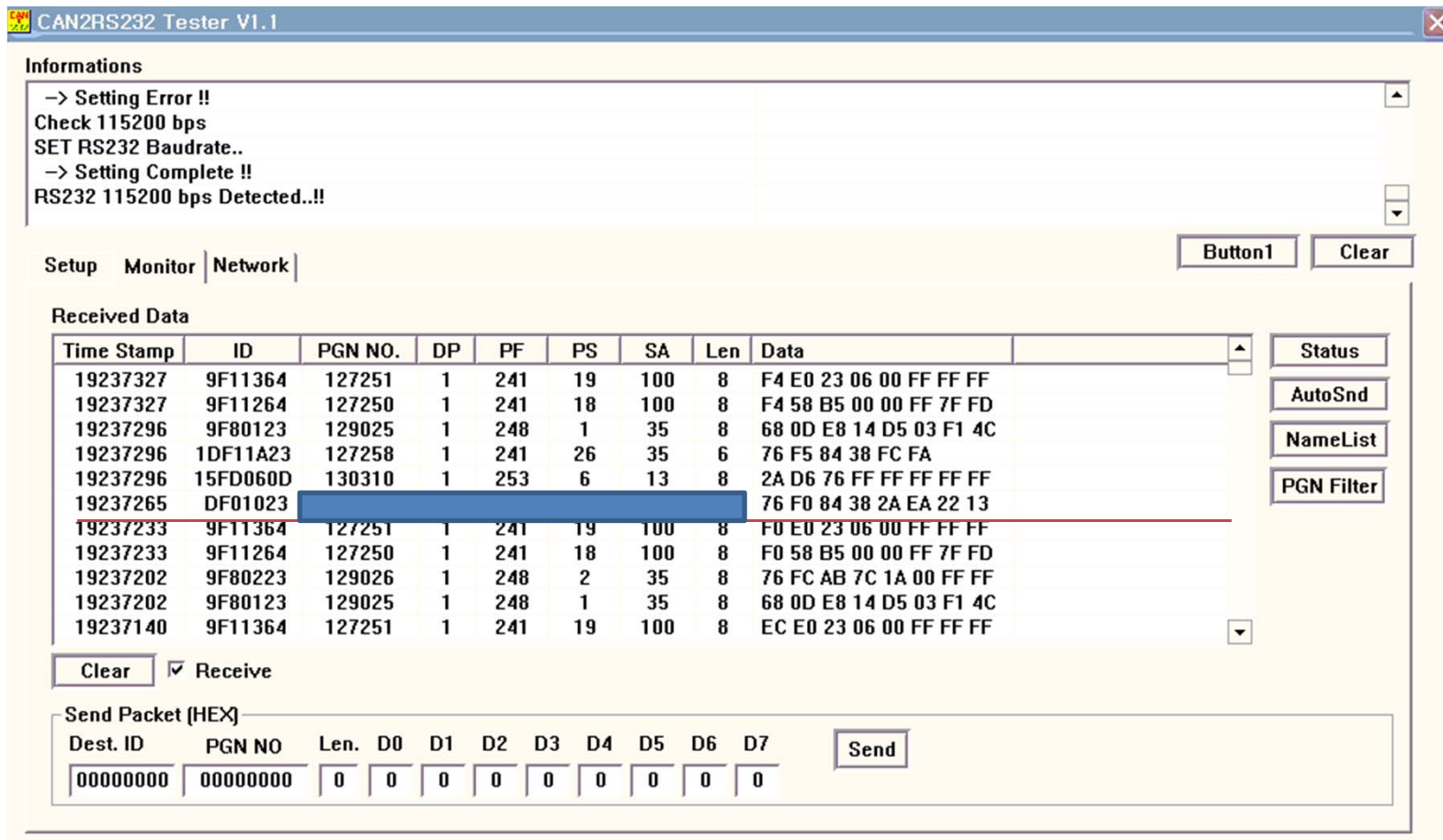
Single Frame: **Yes** Priority Default: **2** Default Update Rate: **100** milliseconds Frequency: **10.** cycles per second
Destination: **Global** Query Support: **Opt'l** ACK Rqmnts:

Field #	Field Name	Byte Field Size	Bit Field Size	Request Parameter	Original Reference ID # 18
1	Latitude	4		No	
	DD022 Latitude, WGS-84		Latitude referenced to WGS-84		
	DF23 Latitude	int32	Range: +/- 90 deg Resolution: 1x10E-7 deg	"-" = South, resolution ~1.1 cm	
2	Longitude	4		No	
	DD023 Longitude, WGS-84		Longitude referenced to WGS-84		
	DF25 Longitude	int32	Range: +/- 180 deg Resolution: 1x10E-7 deg	"-" = West, resolution ~1.1 cm	

Latitude				Longitude			
68	0D	E8	14	D5	03	F1	4C
14	E8	0D	68	4C	F1	03	D5
350752104/10000000 = N35.0752104 deg				1290863573/10000000 = E129.0863573deg			

2. Single Packet 분석

실습과제) 싱글 패킷 분석



2. Single Packet 분석

분석 : DF01023

PDU Format	29 bit Identifier						PGN 개수
	Priority 3bit	EDP 1bit	DP 1bit	PF 8bit	PS 8bit	SA 8bit	
		PGN					
PDU1	0-7	0	0	0-239	DA 0-255	SA 0-255	240
			1	0-239	DA 0-255	SA 0-255	240
PDU2	0-7	0	0	240-255	GE 0-255	SA 0-255	16x256 =4096
			1	240-255	GE 0-255	SA 0-255	16x256 =4096

PDU2:

Priority			EDP	DP	PF								PS							SA								

2. Single Packet 분석

PGN Table : 126992

System Time

PGN: 126992
hex: 1F010

The purpose of this PGN is twofold:

- To provide a regular transmission of UTC time and date

- To provide synchronism for measurement data

This transmission is required to be output on a regular basis with minimal latency to ensure that the SID (sequence identification number) can be used effectively.

Single Frame: Yes

Priority Default: 3

Default Update Rate: 1,000 milliseconds

Frequency: 1. cycles per second

Destination: Global

Query Support: Opt'l

ACK Rqmts:

Field #	Field Name				Original Reference ID # 29
1	SID	Byte Field Size: 1	Bit Field Size:	Request Parameter No	
	DD056 Sequence ID	An upward counting number used to tie related information together between different PGNs. For example, the SID would be used to tie together the COG, SOG and RAIM values to a given position. 255=no valid position fix to tie it to. Range 0 to 250 for valid position fixes.			
	DF53 Integer, 8 bit unsigned	uint8	Range: 0 to 252	Resolution: 1 bit	Unit-less number
2	Source	Byte Field Size:	Bit Field Size: 4	Request Parameter No	
	DD066 Time Source	0x0 = GPS, 0x1 = GLONASS, 0x2 = WWV or equivalent Radio Station Time Sync, 0x3 = Local Cesium clock, 0x4 = Local Rubidium clock, 0x5 = Local Crystal clock, 0x6 - 0xE = reserved, 0xF = Null			
	DF52 Bit field	bit(n)	Range: Variable	Resolution: 1	Used to construct bit fields
3	Reserved Bits	Byte Field Size:	Bit Field Size: resv 4	Request Parameter No	
	DD001 Reserved field	Variable number of reserved bits, all set to logic "1"			
	DF52 Bit field	bit(n)	Range: Variable	Resolution: 1	Used to construct bit fields
4 Bits needed to fill out the byte					
4	Date	Byte Field Size: 2	Bit Field Size:	Request Parameter No	
	DD039 Generic date	Days since January 1, 1970. Date is relative to UTC Time.			
	DF41 Date, day count	uint16	Range: 0 to 65,532 days	Resolution: 1 day	0 = January 1, 1970, max = ~170 years
5	Time	Byte Field Size: 4	Bit Field Size:	Request Parameter No	
	DD158 Generic time of day	24 hour clock. 0 = midnight, time is in UTC			
	DF06 Time of day	uint32	Range: 0 to 86,401 s	Resolution: 1x10E-4 s	~24 hours, 0 = midnight, range allows for up to two leap seconds per day

2. Single Packet 분석

PGN 데이터 분석

Date		Time			

3. ISO Request PGN 59904 실습

예제) Request PGN 59904로 각 디바이스의 60928요청

CAN2USB&RS232 NETWORK MONITORING PROGRAM V1.1

Informations

1. CAN2USB를 선택

2. Open 버튼을 눌러서 close로 바꿈

GATEWAY **CAN2USB** CAN2RS232 Network Packet Monitoring Network Device Monitoring Clear

CAN ID

P(0-7)	EDP(0-1)	DP(0-1)	PF(0-255)	PS(0-255)	SA(0-255)
0	0	0	0	0	0

Sent frames: No. 0 send Message

Baudrate 250 Kbps

Adapters Get Adapters

Available serial numbers LWPNI9M8 >> Serial # to open

Close

Send Message

Msg ID 0x18EA0000

PGN 0 ☐ 59904 ☐ 126208

Flags ☒ Extended ☐ RTR

Send Data 0x00 0xee 0x00 0xff 0xff 0xff 0xff 0xff

Receive Message

Flags ☐ Timestamp ☐ No Local Send ☐ Queue Replace ☐ No auto fetch ☐ Blocking Mode ☐ Slow Mode

Receive Data

Received frames: Clear msg-list

3. ISO Request PGN 59904 실습

예제) Request PGN 59904로 각 디바이스의 60928요청

CAN2USB&RS232 NETWORK MONITORING PROGRAM V1.1

Informations

GATEWAY | CAN2USB | CAN2RS232 | Network Packet Monitoring | Network Device Monitoring | Clear

Received Data

CAN 에서 올라오는 실시간 데이터 리스트

Time Stamp	ID	PGN NO.	DP	PF	PS	SA	Len	Data
647406	9F801F8	129025	1	248	1	248	8	D6 0B E8 14 4D 02 F1 4C
647406	9F11273	127250	1	241	18	115	8	15 FF A1 00 00 FF 7F FD
647328	9F11373	127251	1	241	19	115	8	12 40 6F 0A 00 FF FF FF
647296	9F802F8	129026	1	248	2	248	8	96 FC 2A 35 05 00 FF FF
647296	9F801F8	129025	1	248	1	248	8	D4 0B E8 14 4D 02 F1 4C
647296	9F11273	127250	1	241	18	115	8	11 FF A1 00 00 FF 7F FD
647250	DF11973	127257	1	241	25	115	8	0F FF 7F B5 FF 64 00 FF
647218	9F11373	127251	1	241	19	115	8	0E 40 6F 0A 00 FF FF FF
647203	9F801F8	129025	1	248	1	248	8	D4 0B E8 14 4D 02 F1 4C
647203	9F11273	127250	1	241	18	115	8	0D FF A1 00 00 FF 7F FD
647125	19FA04F8	129540	1	250	4	248	8	AC FF FF FF F1 FF FF FF

Status
AutoSnd
NameList
PGN Filter

3. Device information 업로드 클릭

Clear ☒ Receive 4. Receive 체크박스 클릭

Send Packet (HEX)

Dest. ID	PGN NO	Len.	D0	D1	D2	D3	D4	D5	D6	D7
00000000	00000000	0	0	0	0	0	0	0	0	0

Send

Make Log File
☐ Show Excel
Stop Writing

From: 00h:00m:00s To: 00h:00m:00s Cnt: 0

3. ISO Request PGN 59904 실습

예제) Request PGN 59904로 각 디바이스의 60928요청

PGN 필터 클릭 : 원하는 PGN만 골라서 볼수 있음

5. PGN 필터 클릭

Time Stamp	ID	SA	PGN NO.	Data
1267968	9F11373	127251	1	241 19 115 8 F8 60 B3 0A 00 FF FF FF
1267953	9F11273	127250	1	241 18 115 8 F7 FA A1 00 00 FF 7F FD
1267890	9F802F8	129026	1	248 2 248 8 54 FC F9 59 05 00 FF FF
1267890	9F801F8	129025	1	248 1 248 8 1C 10 E8 14 C5 02 F1 4C
1267875	9F11373	127251	1	248 2 248 8 4D FC AB 2D 06 00 FF FF
1267843	9F11273	127250	1	248 1 248 8 1D 10 E8 14 C4 02 F1 4C
1267781	9F801F8	129025	1	241 19 115 8 E4 00 03 0B 00 FF FF FF
1267493	9F11273	127253	1	241 18 115 8 E3 FA A1 00 00 FF 7F FD
1267390	9F801F8	129025	1	248 1 248 8 1D 10 E8 14 C4 02 F1 4C
1267375	9F11373	127251	1	241 19 115 8 E0 00 03 0B 00 FF FF FF
1267343	9F11273	127250	1	241 18 115 8 DF FA A1 00 00 FF 7F FD

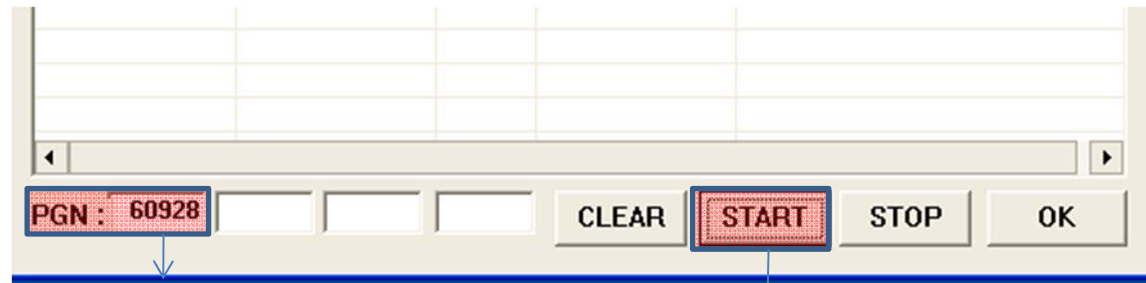
PGN : [] [] [] [] CLEAR START STOP OK

Send Packet (HEX)
Dest. ID PGN NO Len. D0 D1 D2 D3 D4 D5 D6 D7
00000000 00000000 0 0 0 0 0 0 0 0 0 0 Send

Make Log File
Show Excel Stop Writing
From: 00h:00m:00s To: 00h:00m:00s Cnt: 0

3. ISO Request PGN 59904 실습

예제) Request PGN 59904로 각 디바이스의 60928요청

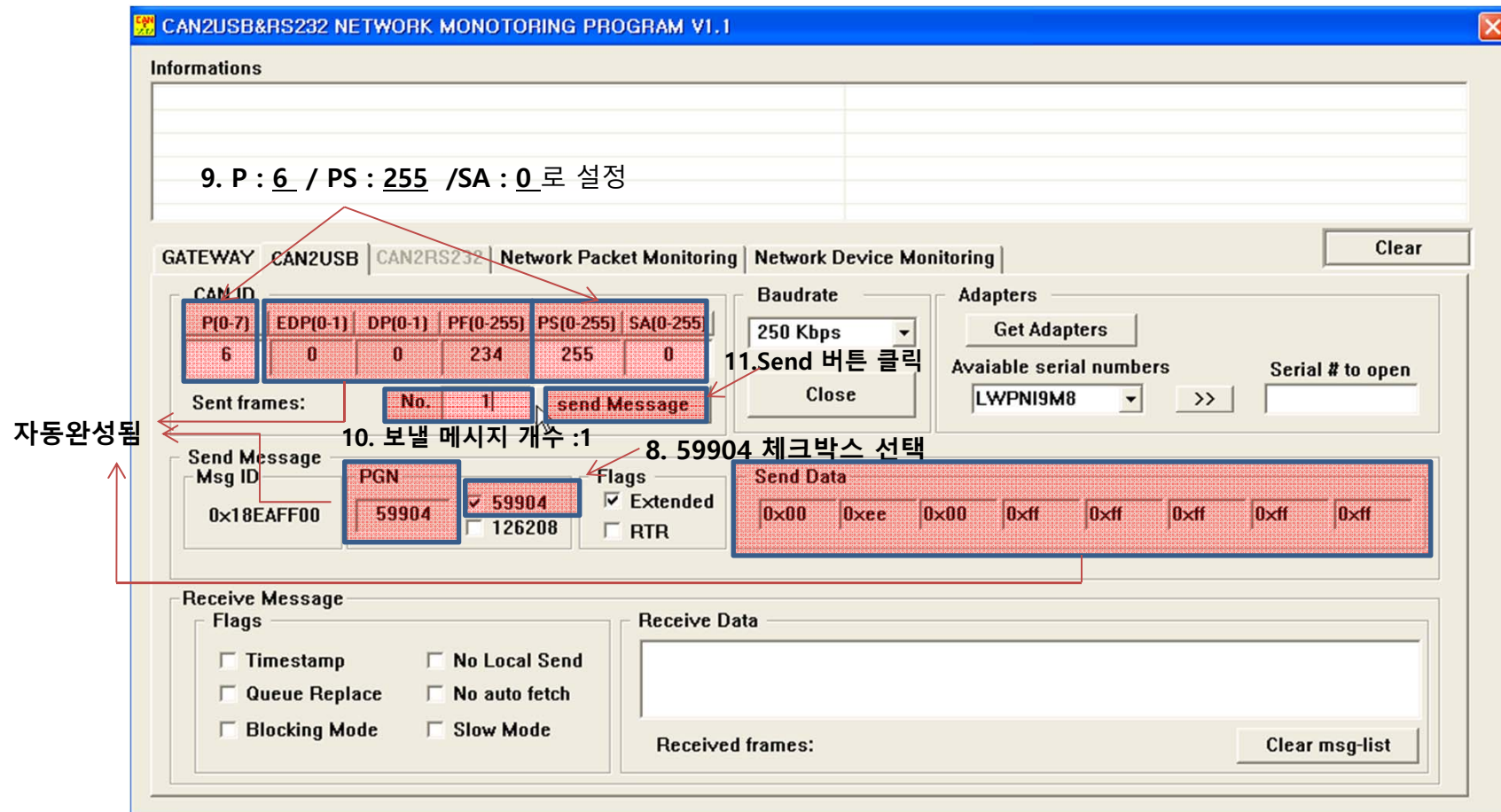


6. 보기 원하는 PGN 입력후 엔터

7. START 버튼 클릭

3. ISO Request PGN 59904 실습

예제) Request PGN 59904로 각 디바이스의 60928요청



3. ISO Request PGN 59904 실습

예제) Request PGN 59904로 각 디바이스의 60928요청

The screenshot displays the CAN2USB&RS232 NETWORK MONITORING PROGRAM V1.1 interface. On the left, the 'PGN Filter' window shows a list of received PGN 60928 data. On the right, the main window shows the 'Network Packet Monitoring' tab with a table of device information.

PGN Filter Data:

Time Stamp	ID	SA	PGN NO.	Data
901562	18EEFFFB	248	60928	4A CA 98 11 00 91 78 C0
901531	18EEFF73	115	60928	8C 1B 31 11 00 A0 50 C0
901515	18EEFF1F	31	60928	A2 25 34 11 00 A0 A0 C0

각 디바이스들로부터 수신된 PGN60928 정보들

Network Packet Monitoring Data:

Manufacturer	Code	Device Class	Code	Function	Code	SA	Ident	Time	ON/OFF
Lowrance Electr...	140	Navigation sys...	60	Global Navigati...	145	248	1624650	2010/8/19/9/53/6	ON
Maretron	137	Steering sys...	40	Heading Sensors	160	115	1121164	2010/8/19/9/53/6	ON
Maretron	137	Instrumentation...	80	General Purpo...	160	31	1320354	2010/8/19/9/53/6	ON

각 디바이스들로부터 수신된 PGN60928의 NAME FIELD를 분석 결과창

PGN : 60928 CLEAR START STOP OK

Check2 DUPLICATED ALLOW Clear

3. ISO Request PGN 59904 실습

실습과제) Request PGN 59904로 각 디바이스의 60928요청 (특정주소: 248,115,31)

CAN2USB&RS232 NETWORK MONITORING PROGRAM V1.1

Informations

PS(PF가 239보다 작을경우엔 DA가 됨)를 248,115,31로 변경해가면서 PGN필터에 나타나는 데이터를 관찰할 것)

GATEWAY CAN2USB CAN2RS232 Network Packet Monitoring Network Device Monitoring Clear

CAN ID

P(0-7)	EDP(0-1)	DP(0-1)	PF(0-255)	PS(0-255)	SA(0-255)
6	0	0	234	255	0

Baudrate: 250 Kbps

Adapters: Get Adapters

Available serial numbers: LWPNI9M8

Serial # to open: >>

Sent frames: No. 1 send Message

Close

Send Message

Msg ID: 0x18EAF00

PGN: 59904

Flags: ☒ 59904 ☐ 126208 ☒ Extended ☐ RTR

Send Data: 0x00 0xee 0x00 0xff 0xff 0xff 0xff 0xff

Receive Message

Flags: ☐ Timestamp ☐ No Local Send ☐ Queue Replace ☐ No auto fetch ☐ Blocking Mode ☐ Slow Mode

Receive Data

Received frames: Clear msg-list

4. ISO Address Claim PGN 60928 분석

예제) PGN 60928 분석

PGN Filter

Time Stamp	ID	SA	PGN NO.	Data
901562	18EFFF8	248	60928	4A CA 98 11 00 91 78 C0
901531	18EFFF3	115	60928	8C 1B 31 11 00 A0 50 C0
901515	18EFFF1F	31	60928	A2 25 34 11 00 A0 A0 C0

각 디바이스들로부터 수신된 PGN60928 정보들

CAN2USB&RS232 NETWORK MONITORING PROGRAM V1.1

Information

GATEWAY | CAN2USB | CAN2RS232 | Network Packet Monitoring | Network Device Monitoring | Clear

Manufacturer	Code	Device Class	Code	Function	Code	SA	Ident	Time	ON_OFF
Lowrance Elect...	140	Navigation sys...	60	Global Navigat...	145	248	1624650	2010/8/19/9/53/6	ON
Maretron	137	Steering syste...	40	Heading Sensors	160	115	1121164	2010/8/19/9/53/6	ON
Maretron	137	Instrumentation...	80	General Purpo...	160	31	1320364	2010/8/19/9/53/6	ON

각 디바이스들로부터 수신된 PGN60928 의 NAME FIELD를 분석 결과창

PGN : 60928 CLEAR START STOP OK

Check2 DUPLICATED ALLOW Clear

PDU1:

Priority			EDP	DP	PF								PS								SA							
1	1	0	0	0	1	1	1	0	1	1	1	0	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0
6			<u>00EE00(16) = 60928(10)</u>										Destination Address(DA):255								Source Address (SA): 248							

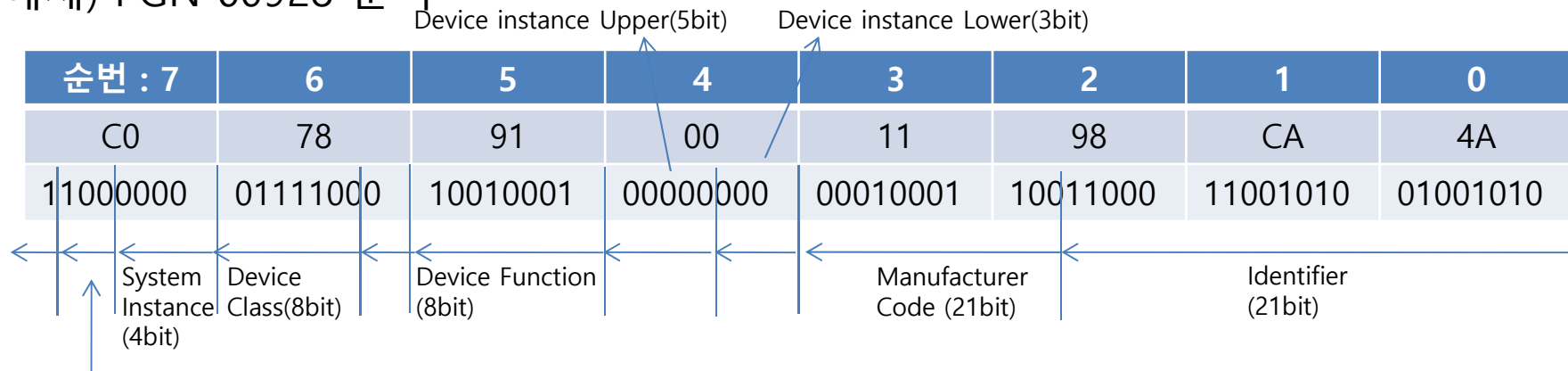
4. ISO Address Claim PGN 60928 분석

예제) PGN 60928 분석

Field #	Field Name	Original Reference ID # 63			
1	Unique Number (ISO Identity Number) DD173 NMEA 2000 Unique Number	Byte Field Size:	Bit Field Size: 21	Request Parameter	No
	DFS2 Bit field	bit(n)	Range: Variable	Resolution: 1	Used to construct bit fields
2	Manufacturer Code DD172 NMEA 2000 Manufacturer Code	Byte Field Size:	Bit Field Size: 11	Request Parameter	Yes
	DFS2 Bit field	bit(n)	Range: Variable	Resolution: 1	Used to construct bit fields
3	Device Instance Lower (ISO ECU Instance) DD201 Generic instance 2 (3-bit)	Byte Field Size:	Bit Field Size: 3	Request Parameter	Yes
	DFS2 Bit field	bit(n)	Range: Variable	Resolution: 1	Used to construct bit fields
4	Device Instance Upper (ISO Function Instance) DD174 Generic instance (5-bit)	Byte Field Size:	Bit Field Size: 5	Request Parameter	Yes
	DFS2 Bit field	bit(n)	Range: Variable	Resolution: 1	Used to construct bit fields
5	Device Function (ISO Function) DD171 NMEA 2000 Function Code	Byte Field Size:	Bit Field Size: 8	Request Parameter	Yes
	DFS2 Bit field	bit(n)	Range: Variable	Resolution: 1	Used to construct bit fields
6	Reserved DD175 Dominant Bit	Byte Field Size:	Bit Field Size: 1	Request Parameter	No
	DFS2 Bit field	bit(n)	Range: Variable	Resolution: 1	Used to construct bit fields
7	Device Class DD170 NMEA 2000 Device Class	Byte Field Size:	Bit Field Size: 7	Request Parameter	Yes
	DFS2 Bit field	bit(n)	Range: Variable	Resolution: 1	Used to construct bit fields
8	System Instance (ISO Device Class Instance) DD169 Generic instance (4-bit)	Byte Field Size:	Bit Field Size: 4	Request Parameter	Yes
	DFS2 Bit field	bit(n)	Range: Variable	Resolution: 1	Used to construct bit fields
9	Industry Group DD168 Industry Group	Byte Field Size:	Bit Field Size: 3	Request Parameter	Yes
	DFS2 Bit field	bit(n)	Range: Variable	Resolution: 1	Used to construct bit fields
10	Reserved (ISO Self Configurable) DD001 Reserved field	Byte Field Size:	Bit Field Size: resv 1	Request Parameter	No
	DFS2 Bit field	bit(n)	Range: Variable	Resolution: 1	Used to construct bit fields

4. ISO Address Claim PGN 60928 분석

예제) PGN 60928 분석



Industry Group (3bit)

Identifier (21bit)

2	1	0
10011000	11001010	01001010
110001100101001001010 (2) = 1624650 (10)		

Manufacturer code (11bit)

3	2
00010001	10011000
00010001100 (2) = 140(10)	
Identifier End	

4. ISO Address Claim PGN 60928 분석

실습과제) PGN 60928 분석

PGN Filter

Time Stamp	ID	SA	PGN NO.	Data
901562	18EEFF8	248	60928	4A CA 98 11 00 91 78 C0
901531	18EEFF73	115	60928	8C 1B 31 11 00 A0 50 C0
901515	18EEFF1F	31	60928	A2 25 34 11 00 A0 A0 C0

각 디바이스들로부터 수신된 PGN60928 정보들

CAN2USB&RS232 NETWORK MONITORING PROGRAM V1.1

Informations

GATEWAY | CAN2USB | CAN2RS232 | Network Packet Monitoring | Network Device Monitoring | Clear

Manufacturer	Code	Device Class	Code	Function	Code	SA	Ident	Time	ON/OFF
Lowrance Electr...	140	Navigation sys...	80	Global Navigati...	145	248	1624650	2010/8/19/9/53/6	ON
Maretron	137	Steering syste...	40	Heading Sensors	160	115	1121164	2010/8/19/9/53/6	ON
Maretron	137	Instrumentation...	80	General Purpo...	160	31	1320364	2010/8/19/9/53/6	ON

각 디바이스들로부터 수신된 PGN60928 의 NAME FIELD를 분석 결과창

PGN : 60928 CLEAR START STOP OK

Check2 Duplicated Allow Clear

PDU1:

Priority			EDP	DP	PF								PS								SA							

4. ISO Address Claim PGN 60928 분석

실습과제) PGN 60928 분석

순번 : 7	6	5	4	3	2	1	0

Identifier (21bit)

Manufacturer code (11bit)

Device Function(8bit)

Device Class(8bit)

5. Address Claiming 실습

예제) 주소충돌 시험

PGN Filter

Time Stamp	ID	SA	PGN NO.	Data
11825578	18EEFF8	248	60928	4A CA 98 11 00 91 78 C0
11825546	18EEFF75	117	60928	8C 1B 31 11 00 A0 50 C0
11825546	18EEFF1F	31	60928	A2 25 34 11 00 A0 A0 C0

주소충돌시험 대상 디바이스(SA:117)

네트워크에 있는 디바이스들

ISO Request PGN59904로 주소요청함

CAN2USB&RS232 NETWORK MONITORING PROGRAM V1.1

Informations

GATEWAY CAN2USB CAN2RS232 Network Packet Monitoring Network Device Monitoring

Clear

BAUDRATE 250 Kbps

Adapters Get Adapters

Available serial numbers LWPNI9M8 >> Serial # to open

Sent frames: No. 1 send Message

Send Message

Msg ID 0x18EAF00 PGN 59904 ☒ 59904 ☐ 126208

Flags ☒ Extended ☐ RTR

Send Data 0x00 0xee 0x00 0xff 0xff 0xff 0xff 0xff

Receive Message

Flags ☐ Timestamp ☐ No Local Send ☐ Queue Replace ☐ No auto fetch ☐ Blocking Mode ☐ Slow Mode

Receive Data

Received frames: Clear msg-list

PGN: 60928 CLEAR START STOP OK

5. Address Claiming 실습

예제) 주소충돌 시험

The screenshot displays the CAN2USB&RS232 NETWORK MONITORING PROGRAM V1.1 interface. On the left, the 'PGN Filter' window shows a table of captured packets. The second packet is highlighted, showing a PGN of 60928 and a Source Address (SA) of 117. A blue arrow points from this packet to the 'SA(0-255)' field in the 'Send Message' section of the main window, which is also set to 117. Another blue arrow points from the 'PGN' field in the 'Send Message' section to the 'PGN' field in the 'PGN Filter' window. The 'Send Message' section also shows the 'Send Data' field with the value 0x8c 0x1b 0x31 0x11 0x00 0x96 0x50 0xc0. The 'Receive Message' section shows the 'Receive Data' field with the value 0x8c 0x1b 0x31 0x11 0x00 0x96 0x50 0xc0. The 'PGN Filter' window has a 'PGN' field set to 60928 and buttons for 'CLEAR', 'START', 'STOP', and 'OK'.

Time Stamp	ID	SA	PGN NO.	Data
12333531	18EEFF76	118	60928	8C 1B 31 11 00 A0 50 C0
12333531	18EEFF75	117	60928	8C 1B 31 11 00 A0 50 C0
12333531	18EEFF75	117	60928	8C 1B 31 11 00 A0 50 C0
12333531	18EEFF75	117	60928	8C 1B 31 11 00 A0 50 C0

PC에서 주소요청 PGN을 네트워크에 보냄
PC의 SA : 117

PC에서 보내는 PGN 데이터 Name field

Received frames: 0

둘다 SA가 117이므로 주소경쟁을 하게됨

5. Address Claiming 실습

예제) 주소충돌 시험

The screenshot shows the CAN2USB&RS232 NETWORK MONITORING PROGRAM V1.1 interface. On the left, the 'PGN Filter' window displays a list of captured packets. The third packet (Time Stamp: 2230812) is highlighted, showing ID: 18EEFF75, SA: 117, PGN NO.: 60928, and Data: 8C 1B 31 11 00 A0 50 C0. A red arrow labeled '1' points from this packet to the 'Send Message' section of the main window. In the main window, the 'Send Message' section shows the PGN set to 60928 and the 'Send Data' field containing the hexadecimal values 0x8c, 0x1b, 0x31, 0x11, 0x00, 0x96, 0x50, and 0xc0. A red arrow labeled '2' points from this 'Send Data' field to the data columns of the table below.

	7	6	5	4	3	2	1	0(순번)
1	C0	50	A0	00	11	31	1B	8C
2	C0	50	96	00	11	31	1B	8C
우선	=	=	A0>96 (2 가 이김)					

비교순서 →

5. Address Claiming 실습

예제) 주소충돌 시험

PGN Filter

Time Stamp	ID	SA	PGN NO.	Data
12333531	18EEFF76	118	60928	8C 1B 31 11 00 A0 50 C0
12230850	18EEFF70	240	60928	41 CA 00 11 00 01 70 C0
12230812	18EEFF75	117	60928	8C 1B 31 11 00 A0 50 C0
12230812	18EEFF1F	31	60928	A2 25 34 11 00 A0 A0 C0

주소경쟁에서 우선순위가 낮은 Heading Sensor 가 자신의 SA를 변경함 (보통 +1)

CAN2USB&RS232 NETWORK MONITORING PROGRAM V1.1

Informations

GATEWAY CAN2USB CAN2RS232 Network Packet Monitoring Network Device Monitoring

Clear

CAN ID

P(0-7) EDP(0-1) DP(0-1) PF(0-255) PS(0-255) SA(0-255)

6 0 0 238 255 117

Baudrate 250 Kbps

Adapters Get Adapters

Sent frames: 2 No. 1 send Message

Close

Available serial numbers LWPNI9M8 Serial # to open

Send Message

Msg ID 0x18EEFF75 PGN 60928 59904 126208

Flags Extended RTR

Send Data 0x8c 0x1b 0x31 0x11 0x00 0x96 0x50 0xc0

Receive Message

Flags Timestamp Queue Replace Blocking Mode No Local Send No auto fetch Slow Mode

Receive Data

-> E id = 18EEFF75 timestamp = 00000000 len = 8 data = 8c 1b 31 11 00 96 50 c0

-> E id = 18EAFF00 timestamp = 00000000 len = 3 data = 00 ee 00

Received frames: 0 Clear msg-list

5. Address Claiming 실습

예제) 주소충돌 시험

이후 최종적으로 변경된 주소를
사용하게 됨

PGN Filter

Time Stamp	ID	SA	PGN NO.	Data
12487452	18EEFF76	240	60928	4A CA 00 11 00 01 78 C0
12487437	18EEFF76	118	60928	8C 1B 31 11 00 A0 50 C0
12487421	18EEFF75	31	60928	A2 25 34 11 00 A0 A0 C0

CAN2USB&RS232 NETWORK MONITORING PROGRAM V1.1

Informations

GATEWAY: CAN2USB | CAN2RS232 | Network Packet Monitoring | Network Device Mo

CAN ID: P(0-7) 6, EDP(0-1) 0, DP(0-1) 0, PF(0-255) 234, PS(0-255) 255, SA(0-255) 0. Baudrate: 250 Kbps. Sent frames: 4, No. 1. send Message. Close.

Send Message: Msg ID 0x18EAF00, PGN 59904 (selected), 59904 (checked), 126208. Flags: Extended (checked), RTR. Send Data: 0x00, 0xee.

Receive Message: Flags: Timestamp, Queue Replace, Blocking Mode, No Local Send, No auto fetch, Slow Mode. Receive Data: -> E id = 18EAF00 timestamp, -> E id = 18EAF00 timestamp, -> E id = 18EEFF75 timestamp. Received frames: 0.

PGN: 60928. CLEAR, START, STOP, OK.

5. Address Claiming 실습

실습과제) 주소충돌 시험

PGN Filter

Time Stamp	ID	SA	PGN NO.	Data
11825578	18EEFF8	248	60928	4A CA 98 11 00 91 78 C0
11825546	18EEFF75	117	60928	8C 1B 31 11 00 A0 50 C0
11825546	18EEFF1F	31	60928	A2 25 34 11 00 A0 A0 C0

주소충돌시험 대상 디바이스(SA:31)

네트워크에 있는 디바이스들

CAN2USB&RS232 NETWORK MONITORING PROGRAM V1.1

ISO Request PGN59904로 주소요청함

GATEWAY CAN2USB CAN2RS232 Network Packet Monitoring Network Device Monitoring

Clear

BAUDRATE: 250 Kbps

Adapters: Get Adapters

Available serial numbers: LWPNI9M8

Serial # to open: >>

Sent frames: No. 1 send Message

Send Message

Msg ID: 0x18EAF00 PGN: 59904

Flags: ☒ Extended ☐ RTR

Send Data: 0x00 0xee 0x00 0xff 0xff 0xff 0xff 0xff

Receive Message

Flags: ☐ Timestamp ☐ Queue Replace ☐ Blocking Mode

☐ No Local Send ☐ No auto fetch ☐ Slow Mode

Receive Data

Received frames: Clear msg-list

PGN: 60928 CLEAR START STOP OK

5. Address Claiming 실습

실습과제) 주소충돌 시험 결과